

Personalized Content Discovery

A Recommendation System on the Netflix Prize Dataset

100M ratings · 480K users · 17,770 movies · 3 models · RMSE + MAP@10

The Problem

Two goals that are not the same goal

Learn preferences from 100,480,507 historical ratings (1-5 stars, 1999-2005)

Predict ratings for movies a user has never seen → measured by RMSE

Rank a Top-10 personalized list per user → measured by MAP@10 (relevant = rating ≥ 3.5)

The catch: a model can predict ratings well and still rank poorly - we evaluate both

Constraint honored: experiments on a reproducible subsample - 8,000 active users $\times$ 4,000 most-rated movies (1.7M ratings)

Why it matters

With 17,770 titles and users rating ~1% of them, the recommender decides which catalog each user ever sees.

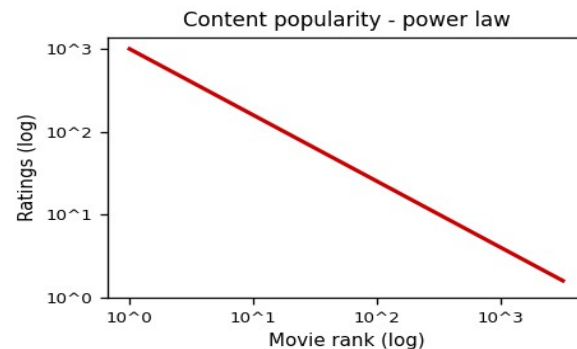
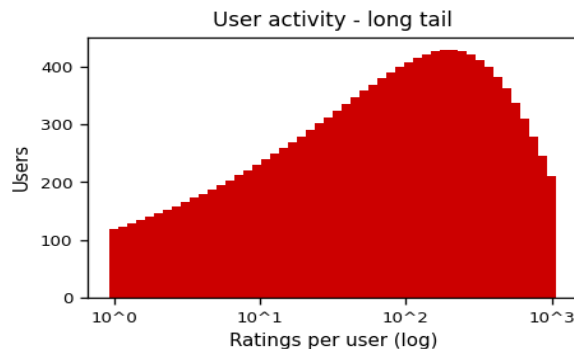
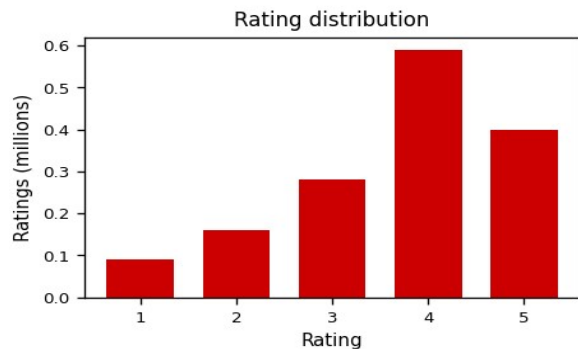
What the Data Says

EDA on 100M ratings - full dataset statistics

Positively skewed: 57% of ratings are 4-5 stars - users self-select what they expect to like

Long-tailed users: median 96 ratings, max 17,653 - design for the quiet majority

Power-law catalog: top 10% of movies = 76.6% of ratings; popularity $\neq$ quality ($r = 0.16$)



Also: ratings drift upward over time (3.41 $\rightarrow$ 3.70) - so we validate temporally, not randomly.

Approach

Three models, one honest comparison

Bias baseline

rating = μ + user bias + item bias

The floor any real model must beat

Item-based CF

adjusted-cosine kNN over co-rated items ($k = 40$)

Interpretable - neighbors are the explanation

Matrix factorization

FunkSVD: 50 latent factors via SGD (numba)

The Netflix Prize workhorse - best accuracy

Evaluation protocol: per-user temporal split (last 20% held out) · RMSE on 336K ratings · MAP@10 over all unseen catalog candidates

Results

RMSE on 336,027 held-out ratings · ranking metrics on 2,000 users

Model	RMSE	MAE	MAP@10	NDCG@10	Coverage	Fit time
Bias baseline	0.917	0.717	0.031	0.058	7.8%	0.1 s
Item-based CF	0.897	0.694	0.036	0.073	67.3%	5.1 s
SVD (k = 50)	0.863	0.669	0.020	0.044	15.7%	34.5 s

SVD wins prediction: RMSE 0.863 - 5.9% better than baseline (Netflix paid \$1M for 10%)

Item-CF wins ranking: MAP@10 +85% vs SVD, recommending from 67% of catalog vs 16%

Headline: the best rating predictor is the worst ranker - optimize for the metric you ship

Recommendations in Practice

SVD scores · item-CF explanations

Success - cult-comedy & drama fan (1,336 ratings)

Loves: Monty Python, Star Wars IV-V, Clerks, The Godfather

Top recommendations:

The Sopranos (S2-S5) 4.2-4.6

Seinfeld S4 · Simpsons S5 · LotR Extended Editions

Explanation shipped with each item:

'Sopranos S2 - because you rated The Godfather 5/5
and The Silence of the Lambs 4/5'

Held-out future watches confirmed the match
(Shawshank, Lost in Translation, Clerks: Uncensored).

Failure - light user (25 ratings)

Profile: Independence Day, Raiders of the Lost Ark,
The American President (mainstream blockbusters)

Got: Sopranos, Lost, Shawshank - generic premium picks

Actually watched: Schindler's List, Indiana Jones 3,
Harry Potter 1 -> clear franchise signals were missed

Diagnosis: sparse-history popularity bias -
the cold-start problem in miniature.

Fix: onboarding ratings, content metadata bootstrap,
history-aware blending toward the bias model.

Key Insights

What we'd tell the ML team

Accuracy $\neq$ ranking. Best-RMSE model ranked worst; an RMSE-only team ships the wrong model.

Coverage is a feature. Item-CF explored 67% of the catalog at better precision; baseline recycled 8%.

Failures are predictable. Quality degrades smoothly with history length - blend by user maturity.

Simplicity is competitive. Two bias terms reach RMSE 0.92; spend engineering on ranking & explanations.

Time matters. Ratings inflated 3.41 $\rightarrow$ 3.70 over the window; random splits would flatter every model.

Explanations are free. Item-CF neighbors double as 'because you watched X' - trust at zero extra cost.

Where this goes next

Ranking-aware training (BPR / WARP) to close SVD's MAP@10 gap

Hybrid content + CF for cold items · time-aware factors for rating drift

Diversity re-ranking (MMR) on the Top-10 · ALS on Spark for the full 100M

Online A/B evaluation - offline MAP@10 is a proxy for engagement, not the answer

Reproducible: github repo · seed 42 · one command per stage - data pipeline, EDA, training, evaluation